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Project Title:	Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays		
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Task Description:	Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes, and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy, and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
	For the grant entitled “Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays” our aim was to study critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following exposure and determine the relationship between these phospho-protein patterns, and two indicators of genomic stability and cancer progression, telomere shortening and centrosome amplification. Sets of mammary epithelial and fibroblast cells from the same individual were used to better understand the etiology of cancer, and whether certain cell types more frequently show evidence of genetic instability upon radiation exposure. By using two cell types from the same individual we were able to determine if cell type was of more importance than genotype in terms of determining the sensitivity to radiation. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, thus telomere length changes may provide a read-out of the carcinogenic process. A second marker we followed after exposure was the frequency of cells showing centrosome amplifications, as these can lead to gains and losses of whole chromosomes and are believed to be due to continuous chromosome mis-segregation during mitosis resulting from these defects. Combinations of increases in number, size, abnormal structure and function characterize centrosome amplification. Centrosome amplification has been detected in a broad range of solid tumors and more recently in leukemia and lymphoma. Centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Cells that over-express pericentrin, a centrosome protein, also exhibit many cellular phenotypes similar to cells from high-grade carcinomas. Therefore centrosome abnormalities are a very good indicator of cancer progression. DNA damage phospho-kinetic signaling as well as these potential markers of carcinogenesis were all followed using flow cytometry-based assays to provide detailed, per cell quantitative data. These studies complemented our earlier studies, and helped us identify unique signaling patterns specific to the time following exposure, the radiation quality, the cell type, and their relationship to carcinogenic endpoints. The radiation quality and track core investigations were performed using a normal hTERT immortalized skin fibroblast line due to the ease in growing this line. Studies were focused on considering the importance of delta-rays when determining the risk of high-LET radiation. Carefully chosen particles of various energies were used to obtain differing populations of cells hit in a greater proportion by tracks or delta-rays to investigate how phosphorylation profiles and carcinogenic endpoints are affected by these components. In total the intent of this work was to provide a comprehensive picture of HZE effects from initial phosphorylation patterns to later phosphorylation profiles, the relationship to various surrogate markers for carcinogenesis, and how track structure affected the signaling response following radiation exposure. NASA areas of interest that were addressed included the linking of two or more cancer development processes and the underlying mechanisms for such endpoints, and their possible inter-relationships. Our study also worked toward improving the understanding of radiation quality effects, establishing the shape of the response curve at low fluences, and the effects on cells directly traversed by HZE nuclei, versus the effect on cells traversed by delta-rays alone. Materials & Methods: Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination of paraformaldehyde and methanol fixation were used depending on what was optimal for the antibody. Fixed slides were sent back to our home institution from Brookhaven in PBS to stain with the appropriate primary and secondary antibodies. Cells exposed in flasks were trypsinized and fixed at various times after exposure, and then sent back on ice in methanol. Aliquots of fixed cells were then stained for various phospho-proteins and analyzed via flow cytometry. For analysis at longer timepoints after exposure (weeks) cells were grown to confluence while at Brookhaven, then frozen down before leaving (~ 1 week post exposure). Cells were then thawed back at the home institution, and expanded to then analyze for telomere length changes and centrosome defects. Cells were stained with pericentrin antibody, then fixed with a methanol fix for flow cytometry based analysis of centrosome dysfunction, and were hybridized to a telomere probe to allow detection of telomere length changes. SUMMARY OF RESEARCH ACCOMPLISHMENTS: Track structure studies based on phospho-kinetics in normal hTERT immortalized fibroblast cells. Normal hTERT immortalized fibroblasts were exposed to a very comprehensive range of carefully chosen high LET ions and energies, to help identify the importance of track structure in defining biological effects. Various phospho-proteins known to be activated following radiation were analyzed for phospho-kinetics including gamma H2AX, pSMC1, and pATF2. Fibroblasts were exposed in triplicate independent experiments to: Fe 300, 600, and 1000 MeV/u; Ti 300, 326, and 386 MeV/u; Si 93, 173, 300, and 1000 MeV/u at timepoints between 0.5 and 24 h. Various comparisons for early and late timepoints were made between equal LET, energy, or Z2/beta2 (roughly equivalent to the number and intensity of delta rays). These studies revealed Z2/beta2 to be a primary determinant in the phosphorylation response. In addition, residual levels of foci (24 h post exposure) correlated well with the predicted energy deposition per nucleosome. This body of work was completed and submitted to Radiation Research recently and is currently in revision (no major revisions

Task Progress:

required). Conclusions from this work include that: 1) most high LET particles induce a greater induction of gamma H2AX and higher level of residual signaling as compared to equal doses of low LET; 2) at early timepoints ATM-mediated phospho-inductions are much lower following all high LET ion exposures as compared to low LET, indicating that the ATM protein is hampered in fully activating down stream proteins following high LET irradiation as compared to low LET exposures; 3) ions with high delta-ray component (Z2/b2) appear to show a greater persistent phosphorylation signal. Phospho-protein kinetics of primary isogenic mammary epithelial and fibroblast cells. To determine how phospho-kinetics may differ based on cell type and status (immortalized vs primary) we compared phosphorylation profiles of gamma H2AX, and pATF2 from primary isogenic (from same tissue biopsy) mammary fibroblasts and mammary epithelial cells. These genetically matched mammary cells where exposed to a variety of different radiation qualities: Fe 300 MeV/u, Ti 300 MeV/ u, Si93 MeV/u, cesium, and X-ray. Initial induction was assessed at 2h post exposure and the amount of residual signaling was assessed at 24 hour post exposure. We chose the 2 h timepoint as for most radiation qualities this timepoint either showed higher levels or the same levels as observed at 0.5 h. Our results indicate that gamma H2AX induction following high LET exposure is greater than following low LET exposure. For example, cells exposed to Si93 MeV/u exhibit a greater induction than following X-ray exposure. Comparing mammary epithelial and fibroblast cells, initial gamma H2AX inductions were similar for all high LET ions, although the Ti ion was uniquely higher for epithelial cells. At late times post exposure, epithelial cells revealed a much greater degree of residual gamma H2AX signal, indicative of a higher amount of unresolved damage in epithelial cells as compared to fibroblasts. This suggests that epithelial cells are more sensitive to high LET damage than fibroblast cells, since the genetic background of the lines cannot be the culprit of this difference as they are from the same individual and tissue section. A second phospho-protein that kinetics were studied in these lines post various radiation quality exposures was pATF2. ATF2 is known to have various functions within the cell however phosphorylation at this particular site (Ser490/498) has been documented to be specifically required for its role in DNA repair. As has been noted in past studies, pATF2 induction was lower than what was observed for gamma H2AX at early times, with levels saturating at about 0.5 Gy. Unexpectedly initial pATF2 induction in fibroblasts was greater than what was observed for epithelial cells for both high and low LET radiation, which is in contrast to the kinetics we observed for gamma H2AX. However, residual signaling for pATF2 was similar for both the fibroblasts and the epithelial cells. Also of interest was the fact that in general for fibroblast line pATF2 levels decreased with time; however, for epithelial cells in many cases the highest levels were observed at the latest timepoint (24h). This is very different from the findings with gamma H2AX, and suggests again that epithelial cells may have more difficulty responding to effects of high LET exposure. The difference observed between the phospho-protein results may also suggest that levels of pATF2 do not directly correlate with the level of double strand breaks (based on gamma H2AX levels), but instead might reflect others things such as perhaps the levels of oxidative stress present in cells. Similarly to the skin fibroblast study currently in revision, we had wanted to analyze data on a per cell basis on exported flow data. We are currently working with a statistician to finalize this work for publishing in next few months. High LET effects on telomere length of primary mammary epithelial and fibroblast cells. To better understand the relationship between radiation exposure and carcinogenesis we analyzed how the phospho-kinetic response following various radiation quality exposures corresponded to the surrogate cancer endpoint of telomere length shortening. The Flow cytometry-based telomere length assay, or flow FISH was used to determine how various qualities of radiation affected telomere lengths of cells differentially. A dose-dependent shortening of telomeres for both mammary epithelial and fibroblast cells following high LET exposures was observed. For low LET exposures, mammary fibroblasts were fairly resistant to shortening effects even at the highest doses, whereas mammary epithelial cells exhibited a dose-dependent shortening even with low LET exposure. The Fe ion irradiations appeared to show the least difference between cell types, in particular Fe 300 MeV/u, which revealed nearly the same change in telomere length over the majority of the doses. Fe 300 MeV/u has the highest delta ray component of all the radiation qualities tested and this may be one of the major drivers in terms of effects, given both cell types failed to show a resistance to the effects. Overall mammary epithelial cells were more sensitive to the effects of radiation and showed a greater decrease in telomere shortening per dose than did fibroblast cells, consistent with the fact that they reveal a greater sensitivity to radiation induced carcinogenesis. We have also exported individual cell flow cytometry data to define how telomere lengths change on an individual cell basis. Preliminary data has revealed that fibroblasts do not show an increase in cells with short telomeres with increasing X-ray dose, yet epithelial cells did reveal an increase in cells showing short telomeres with increasing dose. This additional analysis will allow us to understand in more detail the underlying reasons for the mean changes in telomere shortening for each cell type. Given that mean lengths can change, for example to show an overall longer length, yet the underlying reason for this mean change could be more cells containing shorter telomeres were lost. A finding of an increase in telomere length upon exposure to protons has been noted in the literature, thus detailing the underlying reason for the mean change in shortening should help us to better understand cell type differences and the mechanism involved in radiation-induced telomere shortening. This work is planned to be combined with phospho-kinetic data and as noted above should be submitted in next few months.

High LET effects on centrosome amplification in primary mammary epithelial and fibroblast cells. The second surrogate cancer endpoint, which we have investigated in relationship to radiation quality, dose, and cell type, was centrosome amplifications. Again a flow-based assay was developed and used to identify increases in size or number of centrosomes as a function of dose and radiation quality, using the pericentrin antibody, which envelops the centrosome of the cell. Our preliminary study with mammary epithelial cells following levels days post X-ray exposure was performed to determine the temporal kinetics of centrosome formation and decline following radiation exposure. These studies revealed that at ~day 9 after exposure significant increases in centrosome aberrations were observed as denoted by the percent of cells showing higher pericentrin levels than the 80-90th percentile of controls. Further studies looking at dose and radiation quality effects were then initiated. We have also performed some studies to better understand how flow cytometry results may compare to the more common way of analysis, immunofluorescence and microscopic scoring of aberrant cells. These studies revealed based on Fe 1000MeV/u results, that similar results in terms of pattern of peak levels observed for both the microscopy and FACS assay were at 0.1 Gy. Why an intermediate dose may show higher levels of centrosome aberrations is uncertain; however, it may be that higher doses kill off cells with aberrant levels thus this is the dose where radiation induces damage, yet it is not enough to kill the exposed cells. Binned results for other ions and energies in assays looking at centrosome aberration frequencies were also performed. These studies revealed the greatest percentages of centrosome aberrations were obtained following Fe300 MeV/u exposures for mammary epithelial cells (far right bin for each radiation quality, showing percent of cells having pericentrin levels >80th percentile as compared to controls). Further statistical analysis was performed on individual centrosome cell data to more carefully define differences at lower doses. The Kolmogorov-Smirnov test (K-S test) was used for these studies; it provides a nonparametric test for the equality of a continuous, one-dimensional probability distribution. It was used to investigate how different doses and radiation qualities compared in terms of the level of pericentrin fluorescence (equal to aberrant centrosomes at high levels over controls) following exposure. The K-S test quantifies a distance between the empirical distribution function of the sample and the cumulative distribution function of the control distribution. Based

on KS test we were able to note a significant difference from control levels with all radiation qualities and at all doses for mammary epithelial cells. Results with fibroblast cells from the same tissue (mammary) have shown that fibroblasts are much more resistant to forming centrosome aberrations and that levels in general were below detection using the flow-based assay. We went on to confirm this via the microscopy method following Fe1000 MeV/u exposure, and again found no increase in centrosome aberrations over control levels using this method as well. Thus it appears similarly to the telomere shortening, mammary epithelial cells are more sensitive to the formation of centrosome aberrations following radiation exposure as compared to mammary fibroblasts from the same individual. This again points to a heightened sensitivity for epithelial cells to radiation and potentially also carcinogenesis given these endpoints have been linked to cancer and a pre-malignant state. We are currently completing the analysis of centrosome studies and this work will be compiled into an additional paper which should be out in the summer of 2014.

Conclusions: The findings of this body of work point to the importance of radiation quality and cell type in determining the effectiveness of high LET exposure. Epithelial cells for the most part appear to show a greater sensitivity to high LET exposure and of particular importance also a greater induction in the two surrogate cancer endpoints we investigated, telomere length shortening and centrosome aberration frequencies. Mammary cells, both fibroblast and epithelial cells, appear to show a less robust response to radiation in terms of phosphorylation of various DNA repair proteins after exposures as compared to a skin fibroblast line (82-6). However all lines revealed a greater response to high LET as compared to low LET radiation. These studies allowed us to investigate how well early phospho-protein profiles predict a cell's response to other endpoints more closely aligned to cancer. We determined that there was not a direct correspondence between the ion/energy that showed the highest residual level of double-strand breaks (based on persistent gamma H2AX at 24 h) and the ion/energy that showed the highest level of centrosome defects and/or telomere shortening. For example, for phospho-protein signaling at 24 h Ti300 MeV/u reveals the greatest residual levels for mammary epithelial cells, whereas for centrosome aberrations, Fe 300 MeV/u generates the greatest number of aberrant cells weeks post exposure, and Si 93 MeV/u shows the greatest effect on telomere shortening, again weeks post exposure. However, studies in skin fibroblast cells, with a more comprehensive set of energies for each ion, suggest that the ions/energies that are the most effective when compared to other energies are Ti300, Si93, and Fe300 MeV/u. Thus in each case the ion/energy showing a more dramatic response for our mammary epithelial cells when compared to other energies of the same ion are the one's in which in the skin fibroblast study revealed more effectiveness. Thus, perhaps if additional energies had been used for surrogate cancer endpoints and phospho-kinetics for the mammary cell studies we would have seen a pattern similar to what was observed with the skin fibroblast cells, whereby the ions with the lowest energy, having the greatest delta ray component, were the most effective in each case. Why different types of ions show a preference for inducing greater levels for particular endpoints is uncertain and would require further investigation.

The fact that mammary epithelial cells, rather than mammary fibroblast cells, are more sensitive to radiation based on all the endpoints studied provides support for a greater sensitivity of these cells to radiation-induced carcinogenesis. Further studies using low chronic doses and/or fractionated doses, as well as investigating longer times post exposure would help provide data currently missing in linking exposures more similar to that which astronauts receive, with timepoints more relevant in terms of cancer.

Bibliography Type:	Description: (Last Updated: 11/01/2023)
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